

# MALLESH K

OS KERNEL ENGINEER @ INTEL.

EMAIL:

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Contact No.+91 6363120853

Bangalore (KA)

## Summary

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**20+ years** of professional experienced in Embedded and Linux Developer with excellence history of working in the semiconductors industry. Involved design and development of varies Embedded/Virtualization software's. In-depth experience in Embedded Software, Linux Device Drivers, Linux kernel Debugging, Kernel virtual machine, and embedded/Virtualization Systems. Kernel developer with a passion for performance and scalability.

## Highlights (Experience)

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- **Cache to Cache Optimization** on IA Platform.
- **Enhancing Scheduler optimization** on ESXi.
- Platform Optimization on ESXi operating systems.
- Gen2Gen Optimization on IA platform.
- Enhancing Transport layer on IA platform both ATOM and XEON.
- High performance Next Generation **Control path/Data path/Notification ISA** for Intel Arch.
- Enhanced **Network I/O and Switching** features on Intel IA arch.
- Strong understanding of Kernel internal and data structure such run queues, task\_struct, softirq, kobject etc.
- Knowledge of one or more of the following: **Microprocessor**, Microcontrollers, SoC, device drivers, **Linux Interrupt**, hardware bringup, power management, ARM processors, performance optimization, **x86 ISA/ assembly**, system BIOS or hardware/software integration.
- Exposure on **Kernel network stack internal/NAPI, SKB.**
- Hands on experience of **DPDK Library/drivers and OvS.**
- High degree of proficiency in Linux device driver/system development.
- Good exposure on **Kernel complex issue analysis/root cause and debug.**
- Expert on **Trace32/Lauterbach debugging** using JTAG interface.
- **Linux Systems, low-level & driver** software porting.
- **Linux Multi-Core Programming** on SMP Systems.
- Good hands-on experience with Qualcomm board/Tools such as etc.
- Highly motivated, responsible and quick learner new concepts.
- Good problem solving and troubleshooting skills.
- Requirement gathering & interactions from Customers of different geographical locations.

## Technical Skills

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- Programming Language: Strong **Embedded C**, Assembly, Shell script, Python etc.
- Operating system: Embedded Linux, UNIX, Ubuntu, WinCE, and Palm OS.
- Tools IA/QC/Samsung used: **EMON, EDP, PCM, vTune, Buoy, ITP, CoLIT for Keiko/Coho simulators, QXDM, QPST, QCAT, RDET, Trace32/Lauterbach, GIT, GitHub, GDB, perf, Pktgen**, Virtio Vayu Simulator, Crashescope, **Samsung RDX/Odin 3.81, Power Monitor** etc.
- Linux Kernel/System Programming.

## Work Experiences

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## **OS Kernel Engineer @ Intel, Bangalore KA.**

**-Optimized and enhanced of IA Platform on ESXi Operating system, Oct 2021-Present.**

- Optimized hero features on IA Platform such as SNC, memory latency etc.
- Explored internal of ESXi kernel on IA platform.
- Achieved 10-50% performance improvement on SNC using different workloads.
- Analyzed workload behavior on ESXi scheduler code and listed out feature enhancements.

## **Network Software Engineer @ Intel, Bangalore KA.**

**-Enhancement of ADK for IP Transport and Switching on Snow ridge server, Oct 2018-Sept 2021.**

- Developed Performance framework for ADK on Atom core product.
- Optimized and enhanced Atom/ISA-cldemote, movdir64/software using HW/SW prefetch, cache aware data structure, localities, reduce core snoop/DRAM access etc, Increased L2/L3 occupancy.
- Achieved 2X performance on Single worker by analysis instruction/data flows on IA pipeline engines (FE, EXE, BE) using vTune/emon counters.

## **Architect @ Intel Lab via Wipro, Hillsboro OR.**

**-Secure and Isolation guest communication using next gen. ISA on IA, Jan 2017- Sept 2018.**

- Next generation technology for effective and scalable data transfer between Guests using IA processor.
- Optimized and enhanced ISA/software using HW/SW prefetch, cache aware data structure, localities, reduce core snoop etc.,
- Micro benchmark ISA/Software cost/cycles for coherence/non coherence, cross core snoop, cache align etc. to achieve 2x-5x. Analyzed IA pipeline engines (FE, EXE, BE).

**-Effective and high Scalable vSwitch: Pre-switch, Sept 2015-Dec 2016.**

- Design and implementation of pre-switch kernel module, responsible to evaluate packet switching cost by transferring packet Fast/Slow path entities.
- De-centralized switch cost using offload techniques, in order to scale multiple chain of guests.
- Analysis and Optimization of vSwitch cost/cycles, (N-S, E-W).
- Enhanced I/O devices.

## **Architect @ Motorola/Google via Wipro, Chicago IL.**

**-Kernel development and stability on varies Motorola devices Dec 2014– Aug 2015.**

- Resolved and analyzed varies kernel subsystem issues such Memory management, file system, device driver, network driver, CPU cache error detection etc.
- Debugged using JTAG varies complex HW/SW problems such as NOC error, XPU violation, AHB bus timeout, secure/non-secure watchdog bite and bark.
- Designed and implemented JTAG debugging scripts.
- Identified and resolved ARM CPU/caching bugs and provided kernel workarounds.

## **Technical Lead @ Qualcomm via Sasken Com. Tech, San Diego CA.**

**-Kernel development and stability on Qualcomm Boards/devices. Jan 2013– Nov 2014**

- Bring up of Linux kernel 3.4/3.10 on Snapdragon 8x10 platform, ARMv8.
- General Linux system stability and triaging for Snapdragon MSM8x10/MSM8x26/MSM8994/APQ8084.

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- Pre-silicon Linux bring up and validation on simulator.
- Scripting for automation of Linux validation on emulation platforms.

## **Technical Lead @ TI via Sasken Com. Tech, Dallas Texas.**

### **-Inter processor communication driver for OMAP5. Aug 2012– Dec 2012.**

- Pre-silicon bring up of Linux driver.
- Debugged and Fixed Linux system stability issue using JTAG on Omap5.
- Analyzed Inter processor Linux driver to communicate varies processor cores such as (CortexA15 MPU), DSP, IPU and EVE cores etc.

## **Technical Lead @ Samsung via Sasken Com. Tech, Bangalore India/Seoul S. Korea.**

### **-Kernel development and stability on Samsung devices. Oct 2010 – Jul 2012.**

- Ported Linux device driver on varies Android OS version.
- Debugged and Fixed low level Power issues on Samsung architecture CortexA9MPCORE2SMP (C210).
- General Linux system stability and triaging.

## **Software Eng. @ LnT InfoTech, Bangalore India.**

### **-Tunnel driver for LTE EPC Simulator @ Samsung ODC. Jun 2009 – Sept 2010**

- Design and Implementation of Data bearer connection between SGW and PGW using Linux tunnel driver.
- Debugged and Fixed varies Tunnel driver issues.

### **-Android bring up on OMAP2430 board. June 2008 - May 2009**

- Download initial Android version, Cross compiled on arm and resolved varies issues.
- Varies Linux device driver bring up on OMAP board such touch screen, keypad, RIL, SD card etc.

## **Software Eng. @ Qualcomm (RnD) via LnT InfoTech, San Diego CA.**

### **-Internet access Device. Sep 2006 – May 2008**

- Design and Implementation of OEM static extension for AMSS.
- Using JTAG, analyzed complex feature and enhanced.
- Addressed varies critical issue and fixed based on HID.
- Enabled advanced features for Microbrowser.

## **Team Lead. @ Vistara informatics Pvt Ltd. Bangalore India.**

### **-Math Amigo on Palm Os, Wince. Jun 2004 – Aug 2006**

- Designed and implemented 8, 16 and 32 bit microcontroller software's.
- Ported embedded software to Palm OS, WinCE and desktop.
- Enhanced embedded feature operations.
- Debugged complex embedded issues and fixed those.

## **Publications**

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1. Pushed proprietary kernel patches, dpdk and opnfv patches.
2. [vHost-PCI](#)
3. [Pre-Switch](#)

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4. [DPDK Patches](#)

5. [GitHub](#)

## Key Differentiator

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- ◆ Received three Division Recognition Award @ Intel
- ◆ Received best performer from Samsung Bangalore India.
- ◆ Received Award/Certificate from Sasken, LnT.
- ◆ Developed IA core feature on next generation.
- ◆ Pursued **C and Linux/UNIX kernel internals** Course in Uttara Computers, Bangalore India.

## Academic Background

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| Course           | Branch of Eng.                 | Year | Institution                               | Board/ University | Percentage of marks |
|------------------|--------------------------------|------|-------------------------------------------|-------------------|---------------------|
| BE               | Electronics and Communication. | 2004 | B.V.Bhoomareddy College of Eng. and Tech. | VTU.              | 60%                 |
| Higher Secondary |                                | 1999 | Shivanand college, Kagwad.                | PUC Board         | 65.76%              |
| SSLC             |                                | 1997 | Mallikarjun High school kagwad            | SSLC Board        | 74.76%              |